21 Nitrous Oxide Plant Example (content warning: #explosion)

Decision Guide: Nitrous Oxide Plant Example

Goal

Determine:

- The amount of reactant in the reactor
- Heat generation from reaction
- Reactor temperature behavior
- Whether a thermal runaway or explosion hazard exists

Part 1: Steady-State CSTR Analysis

Step 1: Identify the Type of Problem

Ask yourself:

What reactor?
→ CSTR

Is a reaction occurring?
→ Yes

Is temperature important?
→ Yes

Must an energy balance be written?
→ Yes

This is a non-isothermal reactor problem. Both a mass balance and an energy balance are required.

Step 2: Write the Reaction

The notes indicate the decomposition:

A(l) → B(g) + 2W(g)

representing ammonium nitrate decomposition to nitrous oxide and water.

Step 3: Determine the Conversion

The example provides a conversion close to:

X = 0.9999

or nearly complete conversion.

Use:

$$M_A = M_{A0}(1 - X)$$

to determine the amount of reactant remaining in the reactor.

Step 4: Write the CSTR Mass Balance

At steady state:

$$0 = \dot{m}_{A0} - \dot{m}_A - kM_A$$

or equivalently,

$$M_A = \frac{\dot{m}_{A0} - \dot{m}_A}{k}$$

This allows determination of the reactant inventory in the vessel.

Step 5: Evaluate the Rate Constant

The notes indicate:

k = 0.00517 min^{-1}

or

0.307 hr^{-1}

at approximately T = 510°F. If temperature changes, an Arrhenius expression must be used.

Part 2: Energy Balance

Step 6: Identify Heat Effects

The reaction is strongly exothermic. Therefore:

Reaction
$\downarrow$
Generates Heat

while feed streams and cooling remove energy.

Step 7: Write the Energy Balance

General form: $\text{Accumulation} = \text{In} - \text{Out} + \text{Generation} + Q + W_s$

For steady-state operation: $0 \; = \text{In} - \text{Out} + \text{Generation} + Q + W_s$

where:

- Q = heat added to or removed from the reactor
- W_s = shaft work

This is an adiabatic reactor, so Q = 0.

Step 8: Account for Latent and Sensible Heat

The feed and product streams enter and leave at different phases and temperatures, so latent and sensible-heat effects must be included in the energy balance.

Step 9: Include the Heat of Reaction

The dominant heat-generation term is:

$$\dot{m}_{A0} X(-\Delta H_{\text{rxn}})$$

where:

- $\dot{m}_{\text{A0}}$ = feed rate of reactant A

- X = conversion
- ΔH_{rxn} = heat of reaction

This term represents the total heat released by the chemical reaction.

Physical Interpretation

More Conversion
↓
More Reaction
↓
More Heat Released

Because this decomposition reaction is highly exothermic, the heat-of-reaction term is often much larger than the sensible-heat corrections and therefore dominates the thermal behavior of the reactor.

The notes indicate that heat-capacity corrections are small compared with the reaction enthalpy, so the heat of reaction may be approximated as constant.

Step 10: Identify the Source of Heat Removal

During normal operation, the reactor appears to be adiabatic:

$Q = 0$

However, the reactor is **not truly thermally isolated**. The incoming feed provides a significant heat sink because a portion of the reaction heat is consumed by Heating the feed + Vaporizing the feed. The latent heat required to convert liquid feed into vapor continuously removes energy from the reactor.

Physical Interpretation

Reaction Releases Heat
↓
Feed Evaporates
↓
Energy Consumed

↓
Temperature Controlled

As long as feed continues entering the reactor, vaporization acts as an effective cooling mechanism.

What Happens if the Feed Is Shut Off?

When the feed is interrupted:

No Fresh Feed
↓
No Vaporization
↓
No Latent-Heat Removal
↓
Loss of Cooling

The reactor now behaves much more like an **adiabatic batch reactor**. Heat generation from the decomposition reaction continues, but the primary mechanism that had been absorbing that heat disappears.

Why Is This Dangerous?

The decomposition reaction is highly exothermic. After feed shutoff:

Reaction Continues
↓
Heat Continues to Be Generated
↓
Temperature Increases
↓
Reaction Rate Increases
↓
More Heat Generated

This creates the conditions for thermal runaway.

Key Insight

Normal Operation

Reaction Heat
↓
Consumed by Feed Vaporization
↓
Stable Temperature

Feed Shutoff

No Vaporization
↓
No Heat Removal
↓
Adiabatic Behavior
↓
Temperature Rise
↓
Potential Runaway

This is the critical transition in the analysis: **turning off the feed removes the reactor's primary cooling mechanism, which converts a stable operating reactor into an adiabatic runaway scenario.**

Part 3: Thermal Runaway Analysis

Step 11: Model as an Adiabatic Batch Reactor

The notes then suggest to model like an adiabatic batch reactor for a feed-shutoff or upset scenario. This is where runaway conditions are investigated.

Step 12: Write the Batch Conversion Equation

Use:

$$\frac{dX}{dt} = k(1 - X)$$

The amount of reactant remaining in the reactor is related to conversion by:

$$M_{\mathrm{A}} = M_{\mathrm{A0}}(1 - X)$$

where:

- M_{A} = mass of reactant A remaining in the reactor
- M_{A0} = initial mass of reactant A
- X = conversion

Physical Interpretation

Higher Conversion
↓
Less Reactant Remaining

When $X = 0$ all of the reactant remains in the reactor. When $X = 1$ all of the reactant has been consumed.

Step 13: Include Temperature-Dependent Kinetics

The rate constant varies with temperature according to the Arrhenius equation:

$$k(T) = k(T_{ref})\left[\frac{E}{R}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right]$$

where:

- k = rate constant at temperature T
- E = activation energy
- R = gas constant
- T_{ref} = reference temperature
- T = reactor temperature

Step 14: Write the Adiabatic Energy Balance

The notes show a relationship of the form:

$$T = T_0 + \frac{X(-\Delta H_{rxn})}{\sum C_p}$$

which indicates:

More Conversion
↓
More Heat Release
↓
Higher Temperature

Step 15: Recognize the Runaway Feedback Loop

Reaction Generates Heat
↓
Temperature Increases
↓
Rate Constant Increases
↓
Reaction Speeds Up
↓
More Heat Generated

This positive feedback mechanism is the reason decomposition reactors can experience thermal runaway and explosion hazards.

Final Answer Checklist

- ☑ Wrote the CSTR mass balance.
- ☑ Calculated the reactant inventory.
- ☑ Wrote the energy balance.
- ☑ Included latent and sensible-heat effects.
- ☑ Included the heat of reaction.
- ☑ Developed the adiabatic upset model.
- ☑ Evaluated the potential for thermal runaway.

CBE 3610 Reactor Design Handbook Equations Needed

Eq. (15) CSTR Mole Balance

Eq. (20) Batch Reactor Mole Balance

Eq. (41) Batch Reactor Energy Balance

Eq. (49) Arrhenius Equation

Key Insight

Normal Operation
↓
Mass Balance
+
Energy Balance
↓
Safe Temperature

Turn off feed = Loss of Cooling
↓
Adiabatic Behavior
↓
Temperature Rise
↓
Faster Reaction
↓
Potential Runaway

The central lesson of this example is that highly exothermic decomposition reactions can be stable during normal operation but become dangerous during upset conditions because temperature and reaction rate reinforce one another.